

Experiment 10: Predictive Analytics to Analyze Microarray Data

Aim

To perform predictive analytics by selecting important features using statistical methods and building a classification model using Random Forest to evaluate performance on a biomedical dataset.

Code

```
import numpy as np
import pandas as pd

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.feature_selection import SelectKBest, f_classif
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

# -----
# STEP 1: Load Built-in Dataset
# -----
data = load_breast_cancer()

X = pd.DataFrame(data.data, columns=data.feature_names)
y = pd.Series(data.target)

print("Dataset shape:", X.shape)

# -----
# STEP 2: Normalize Data
# -----
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```

# -----
# STEP 3: Feature Selection
# -----

k = 10 # select top 10 features (like selecting important genes)
selector = SelectKBest(score_func=f_classif, k=k)
X_selected = selector.fit_transform(X_scaled, y)

print("Selected feature shape:", X_selected.shape)

# -----
# STEP 4: Train-Test Split
# -----

X_train, X_test, y_train, y_test = train_test_split(
    X_selected, y, test_size=0.2, random_state=42
)

# -----
# STEP 5: Train Model
# -----

model = RandomForestClassifier(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# -----
# STEP 6: Prediction
# -----

y_pred = model.predict(X_test)

# -----
# STEP 7: Evaluation
# -----

print("Accuracy:", accuracy_score(y_test, y_pred))

```

```
print("\nClassification Report:\n", classification_report(y_test, y_pred))
print("\nConfusion Matrix:\n", confusion_matrix(y_test, y_pred))
```

Output

Dataset shape: (569, 30)
Selected feature shape: (569, 10)
Accuracy: 0.956140350877193

Classification Report:

	precision	recall	f1-score	support
0	0.95	0.93	0.94	43
1	0.96	0.97	0.97	71
accuracy				0.96 114
macro avg				0.96 0.95 0.95 114
weighted avg				0.96 0.96 0.96 114

Confusion Matrix:

```
[[40 3]
 [ 2 69]]
```

Result

The dataset was normalized and important features were selected using statistical methods. A Random Forest classifier was trained on the selected features to perform prediction. The model achieved good accuracy, and evaluation metrics such as classification report and confusion matrix demonstrated its effectiveness in predictive analytics for classification tasks.